

BEDANTA BANDYOPADHYAY

Electronics Engineer | Software Developer | Full-Stack & Cloud Engineering | ML / Embedded Systems

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SUMMARY

Software developer building full-stack, cloud-deployed applications, with a B.Tech in Electronics & Communication Engineering (CGPA 8.86) and an active data-structures-and-algorithms practice. Shipped production-style systems spanning React/Node.js applications, PostgreSQL and Redis-backed backends with real-time pub-sub, cascaded LLM/AI integrations, and IoT platforms deployed on AWS, Render, and GitHub Pages. Pairs rigorous problem-solving and ongoing competitive programming with a 7+ year record of independently managing client-facing technical projects as a freelance CGI/3D engineer. Seeking software engineering roles where systems thinking, fast execution, and end-to-end ownership matter.

EDUCATION

B.Tech, Electronics & Communication Engineering — Institute of Engineering & Management | Expected 2027 | CGPA: 8.86

Higher Secondary (ISC) — Ram Mohan Mission High School | 2023 | Percentage: 73.2%

Secondary (ICSE) — Ram Mohan Mission High School | 2021 | Percentage: 94.5%

TECHNICAL SKILLS

Languages: Python, C/C++, Java, JavaScript, Rust, SQL

Web & Backend: React.js, Node.js, Express.js, Flask, REST API design, HTML/CSS

Databases & Real-Time: PostgreSQL, MongoDB, Redis, LISTEN/NOTIFY pub-sub, Server-Sent Events (SSE)

Cloud, Auth & DevOps: AWS, Firebase (Auth/Firestore), CI/CD pipelines, Render, GitHub Pages, Git/GitHub, RBAC, Docker, JWT

AI / Machine Learning: Scikit-learn, LangChain, RAG, Hugging Face, Deep Learning (CNNs), Computer Vision, LLM API integration

Data Structures & Algorithms: Competitive programming (arrays, graphs, DP, prefix sums, BFS, tree algorithms, Euler tour/LCA)

Embedded & IoT: Arduino, Raspberry Pi, Jetson, KiCAD, Cadence Virtuoso, LTSpice, HFSS, MATLAB, IoT sensor integration

CAD & 3D (freelance domain): AutoCAD, SolidWorks, Fusion 360, Blender3D, Maya, 3ds Max, V-Ray/Corona/KeyShot rendering

PROJECTS

1. CodeJudge — Browser-Based Coding Classroom & Auto-Grading Platform | Jul '26

- Built a Dockerized, multi-language (Python/C/C++/Java) code-execution sandbox with ulimit-capped resources and privilege-dropped execution, giving tuition students without personal computers a secure browser-based IDE to write and run code from any device.
- Implemented a PostgreSQL-backed assignment judge with deadline-gated access, hidden test cases, and LeetCode-style verdicts, plus JWT/RBAC auth and automated student onboarding via a Google Form webhook and Resend transactional email.

Tech: React, Node.js, Express.js, PostgreSQL, Docker, JWT, Resend, Render, GitHub Pages

Link: pandadsqn.github.io/CodeJudge

2. Infrastructure-Assassin — AI-Driven Infrastructure Testing & Resilience Suite | Jul '26

- Built a multi-tier, RBAC-governed platform using PostgreSQL for persistent telemetry and Redis for session caching, with LISTEN/NOTIFY pub-sub over Server-Sent Events for real-time dashboard updates.
- Integrated a cascaded LLM pipeline (Gemini API, Ollama, Grok cascaded) for automated threat analysis, added Firebase Authentication, and shipped CI/CD to a multi-instance Render backend + GitHub Pages frontend.

Tech: React, Node.js, PostgreSQL, Redis, Firebase, Gemini API, CI/CD, Render, GitHub Pages

Link: pandadsqn.github.io/Infrastructure-Assassin-Production

3. OccuNova — Deep Learning Diagnostic System for Glaucoma Detection | Jun '26

- Designed a cascaded ResNet50 computer-vision pipeline with AI-driven UNet cropping to isolate optic-nerve regions from retinal fundus scans.
- Built a scalable inference backend and React/Node.js frontend covering the full lifecycle from clinical data preprocessing to model training and deployment.

Tech: Python, CNNs, Computer Vision, Node.js, React, Git/GitHub

Link: pandadsqn.github.io/OccuNova

4. Agro-Guard — IoT Crop & Grain-Silo Monitoring System | Feb '26

- Engineered an Arduino-based sensor pipeline (DHT11, HC-SR04, LDR) for continuous environmental telemetry capture in agricultural storage.
- Built AI-driven analytics for grain storage management on a scalable backend deployed on Render, spanning hardware integration through cloud deployment.

Tech: Arduino, IoT sensors, Node.js, React, Render, Git/GitHub

Link: pandadsqn.github.io/Agro-Guard

PROFESSIONAL EXPERIENCE

1. 3D Generalist & Environment Artist — The AutoFocus Studio | Aug '25 – Feb '26

- Delivered high-fidelity visualizations and technical artwork for studio projects and clients across environment design and generalist 3D workflows.
- Worked on the launch campaign for the Hyundai Creta N Line Knight Edition, creating 360° exterior imagery for the website.

2. Co-Lead — Ninja Fellowship | Nov '22 – Mar '25

- Led a team of 3D artists over 2+ years, coordinating delivery of visual assets for multiple brand engagements.

3. Automotive & Product Visualizer (Freelance) | Apr '19 – Aug '25

- 7+ years independently managing end-to-end client projects, delivering hyper-accurate 3D design and lighting for automotive and product visualization.
- Collaborated with PJKT on a classified project built on a design by Sasha Selipanov (former Bugatti / Koenigsegg designer).

INTERNSHIP & CERTIFICATIONS

Advanced & Cleaner Materials in Solar Technology — Internship at IEM Newtown, IEDC | Jun '25 – Jul '25

- Optimized performance of tri-layer, lead-free, carbon-based perovskite solar cells using machine learning; authored a full technical report.

Certifications:

- Business Intelligence & Analytics — NPTEL, IIT Madras (Jun '25)
- Cloud Computing — NPTEL, IIT Kharagpur (Jun '25)
- Joy of Computing with Python — NPTEL, IIT Ropar (Nov '24)
- Introduction to IoT — Coursera, UC San Diego (Oct '24)

PUBLICATION

AgroGuard: A Novel IoT Enabled Framework for Smart Crop Preservation in Grain Silos

Publisher — Bentham Science Journal

Status — Ongoing

LEADERSHIP & EXTRACURRICULAR

- Computer Society of India – UEM: Vice President (Jan '26–Present), Manager (Aug '25–Jan '26), Graphics Lead (Mar '25–Jan '26)
- Iris, UEMK Photography Society: Tech/Media/Graphics Lead (Sep '24–Aug '25)
- Class Representative (Jan '26–Present)